

Ben-Gurion University of the Negev

The Faculty of Engineering

The Department of Industrial Engineering and Management

A Thesis Template for Ben-Gurion University

Thesis submitted in partial fulfillment of the requirements
for the degree of Doctor of Philosophy

Israel Israeli

Under the supervision of **Prof. Max Musterman**

September 2026



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Signature of student: _____ Date: _____

Signature of supervisor: _____ Date: _____

Signature of chairperson of the
committee for graduate studies: _____ Date: _____

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Abstract

This document is a template for a thesis at Ben-Gurion University of the Negev, and at the same time its manual. An English thesis and the Hebrew summary, abstract and cover that the university requires at the end of the bound volume are produced as one PDF whose links work in both directions: the English table of contents lists the Hebrew pages, the Hebrew table of contents shows the English chapter numbers and pages read from the English build, and Hebrew text can cite the bibliography and link to English theorems, figures and definitions while printing their numbers automatically.

The chapters explain the build process, the anchor and label conventions that connect the two parts, and the commands for writing Hebrew, and they exercise every construct a thesis needs along the way: numbered chapters and sections, a theorem stated in the introduction and restated with its proof, a definition, a figure, a table, an algorithm, code listings and citations.

Replace this abstract with your own.

Acknowledgements

I thank my supervisor, Prof. Max Musterman, for guidance, patience, and the occasional well-timed question. I thank the members of my committee, my colleagues at The Department of Industrial Engineering and Management, and the people who read drafts of this thesis.

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Chapter 1

Introduction

1.1 What This Template Gives You

A thesis at Ben-Gurion University is written in English and bound with a Hebrew summary, a Hebrew abstract and a Hebrew cover at its end, read from the back. LaTeX cannot typeset both halves in one document without pain, so this template keeps them apart: `main.tex` is the English thesis, compiled with pdfLaTeX, and `hebrew_pages.tex` is the Hebrew part, compiled with XeLaTeX. A script compiles both and merges them into one PDF, the one you are reading, in which

- the English table of contents lists the Hebrew pages with their real page numbers,
- the Hebrew table of contents shows the English chapter and section numbers and pages, read automatically from the English build, and each line of it is a link into the English part,
- Hebrew text can cite the bibliography and refer to any English theorem, figure or definition by a link that also prints its number,
- page numbers run continuously through both parts, and
- every link keeps working after the PDF has been opened and re-saved by any viewer.

This document is both the demonstration and the manual. Chapter 2 explains the build, Chapter 3 the anchors that connect the two parts, and Chapter 4 the commands for writing Hebrew. Along the way, every construct a thesis needs appears at least once: a theorem stated here and restated with its proof later, a definition, a figure, a table, an algorithm, code listings, citations such as [Kar72, GJ79] or [DF13, Cyg+15], and cross-references with `\Cref`.

The whole idea of the template fits in one statement, which Chapter 3 proves.

Theorem 1.1 *Every object in the English part that carries a named anchor can be linked to from the Hebrew pages, and the link survives any PDF viewer.*

1.2 Quick Start

1. Fill in `thesis_info.tex`: title, author, supervisor, department, degree and date, in English and Hebrew. Both documents read this file, so the three cover pages and the abstracts agree.

2. Write your chapters in `content/` and list them in `main.tex` by copying the chapter block shown in Section 3.2.

3. Write the Hebrew summary and abstract in `hebrew_pages.tex`, and mirror your chapters in its table of contents (Section 4.4).

4. Run `./compile_thesis.sh` and open `build/thesis.pdf`. The merge step must print no warning (Section 3.4).

Table 1.1 lists the files of the template.

Table 1.1: The files of the template.

File	Role
`thesis_info.tex`	title, names, department, degree, date (English and Hebrew)
`main.tex`	the English document: preamble, front matter, chapter list, bibliography
`content/*.tex`	one file per chapter
`Main_pages/`	cover, signature page, abstract, acknowledgements, affidavit, logo
`hebrew_pages.tex`	the Hebrew document: Hebrew TOC, summary, abstract, Hebrew cover
`bibliography/references.bib`	the bibliography, shared by both documents
`compile_thesis.sh`	builds everything into `build/thesis.pdf`
`merge_thesis.py`	concatenates the two PDFs and rewires the links

Chapter 2

The Build Process

2.1 What the Script Does

`compile_thesis.sh` needs TeX Live (or MacTeX) with `latexmk`, `pdflatex`, `xelatex` and `biber`, and Python 3 with the `pikepdf` package. It runs the three steps of Figure 2.1, all inside the git-ignored folder `build/`.

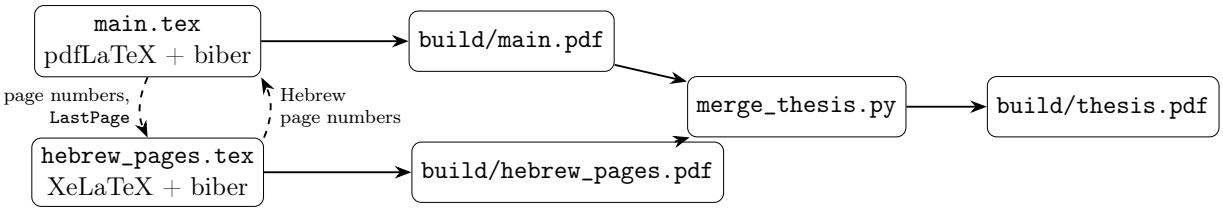


Figure 2.1: The build. Solid arrows are the three steps of the script; dashed arrows are the labels each document reads from the other's auxiliary file, which is why the first two steps are repeated until both settle.

1. `latexmk -pdf main.tex` produces `build/main.pdf`. `latexmk` runs biber and repeats pdfLaTeX until the references are stable.
2. `latexmk -xelatex hebrew_pages.tex` produces `build/hebrew_pages.pdf`.
3. `merge_thesis.py` appends the Hebrew pages to the English PDF and rewires the cross-document links (Section 3.4). The result is `build/thesis.pdf`.

`./compile_thesis.sh clean` deletes `build/`. A clean build takes a few minutes; later builds only redo what changed.

2.2 Why the First Two Steps Repeat

Each document reads labels from the other. The Hebrew document continues the page numbering after the English `LastPage` label and prints the English chapter numbers and

pages in its table of contents. The English document, in turn, needs the page numbers of the Hebrew summary and abstract for its own table of contents. Both documents load the `xr-hyper` package, which imports the labels of another document's `.aux` file under a prefix: `en-` for English labels read by the Hebrew document, `he-` for Hebrew labels read by the English one.

Because each side depends on the other, the script compiles English, then Hebrew, then checks whether either `.aux` file changed, and repeats the pair until neither does. On a clean build the first English pass prints page 0 for the Hebrew entries, since no Hebrew `.aux` exists yet; the second pass fixes it. From the second pass on, the script passes `-g` to `latexmk`, because `latexmk` does not know that the other document's `.aux` file is a dependency and would otherwise report that there is nothing to do.

2.3 Fonts, Overleaf and Troubleshooting

2.3.0.0.1 Fonts. The Hebrew pages use *Linux Libertine O* for Latin text, *David CLM* for Hebrew and *Latin Modern Roman* for Latin fragments inside Hebrew paragraphs. All three ship with TeX Live (packages `libertine`, `culmus`, `lm`), but XeTeX on macOS only finds a font by name if macOS itself knows it. If the build stops with `The font "Linux Libertine O" cannot be found`, run `./compile_thesis.sh install-fonts` once; it copies the font files into `~/Library/Fonts`. On Linux and on Overleaf this is not needed.

2.3.0.0.2 Overleaf. Overleaf cannot run the merge script. Compile `main.tex` with pdfLaTeX and `hebrew_pages.tex` with XeLaTeX (two projects, or one project with the main file switched), download both PDFs and merge locally with `python3 merge_thesis.py english.pdf hebrew.pdf out.pdf`.

2.3.0.0.3 Symptoms and causes.

- *Hebrew Summary 0 in the English TOC*: the English part was not rebuilt after the Hebrew part. Run the script again rather than `latexmk` by hand.
- *Merge warning about names that exist in neither PDF*: a Hebrew `\hyperlink` has no matching anchor in the English source. See Section 3.4.
- *Nonsense errors such as `Missing \begin{document}` in `build/main.aux`*: a build was interrupted or two ran at once and an `.aux` file is truncated. Run `./compile_thesis.sh clean` and build again.
- *biber: extracting arm64 binary with lipo failed* (macOS, after a system update): the universal `biber` shipped with MacTeX unpacks itself with `lipo`, which newer command line tools reject. Extract the slice once by hand, `lipo -thin arm64 $(which biber) -output ~/bin/biber`, and put `~/bin` first on your `PATH`.

- *Package bidi error about xcolor*: in `hebrew_pages.tex`, any package that loads `xcolor` must come before `bidi` and `polyglossia`. The template loads `xcolor` first for this reason.

Chapter 3

Anchors and Links Between the Two Parts

3.1 Anchors and Labels

Two different things connect the Hebrew pages to the English ones, and it helps to keep them apart.

Definition 1 *An* anchor *is a named position in a PDF, created by* `\hypertarget{name}{}` *and reached by* `\hyperlink{name}{text}`. *A* label *is a LaTeX name, created by* `\label{name}`*, from which* `\ref` *and* `\pageref` *print a number and a page.*

Anchors are what a click follows; labels are what gets printed. The Hebrew document can use both across the document boundary: labels through `xr-hyper`, and anchors through the merge step, since an anchor's name is stored in the PDF. The template's convention is to give every object the Hebrew pages may refer to both an anchor and a label, named as in Table 3.1.

Table 3.1: Naming conventions. The Hebrew document reads labels with the prefix `en-`.

Object	Anchor (for `\hyperlink`)	Label (for `\ref*`, `\pageref*`)
Chapter	`chapter:<key>`	`chapter:<key>`
Section	`target:sec:<key>`	`toc:sec:<key>`
Theorem, figure, table, definition	`target:thm:<key>`, `target:fig:<key>`, ...	`thm:<key>`, `fig:<key>`, ...
Bibliography	`chapter:bib`	`toc:bibliography`
Bibliography entry	its bib key	
Hebrew summary, abstract	`chapter:hebrew:summary`, `chapter:hebrew:abstract`	`toc:hebrew:summary`, `toc:hebrew:abstract`

3.2 Chapters and Sections

Every chapter in `main.tex` follows the same block. The `\phantomsection` gives hyperref a correct bookmark, the `\hypertarget` is the anchor the Hebrew TOC links to, and the `\label` is where it reads the chapter number and page.

```
\cleardoublepage
\phantomsection
\hypertarget{chapter:build}{}%
\chapter{The Build Process}
\label{chapter:build}
\input{content/build}
```

Sections use the helper `\linkedsection`, defined in `main.tex`:

```
\linkedsection{script}{What the Script Does}
% expands to
\section[What the Script Does]{\hypertarget{target:sec:script}{}What the Script
    Does}
\label{toc:sec:script}
```

The anchor is placed *inside* the heading on purpose. An anchor written on the line before `\section` is set at the end of the preceding paragraph, so whenever the section happens to start a new page, a click lands at the bottom of the previous one. Add your own `\label{sec:...}` after `\linkedsection` for `\Cref`, as this section does.

An unnumbered chapter, such as the conclusion of this document, has no number to read, so it needs only the anchor and a label, plus `\addcontentsline` to appear in the English TOC. Algorithm 3.1 summarises the steps for adding a chapter that both tables of contents can see.

Algorithm 3.1: Adding a chapter

Input: a chapter title, a short key, a Hebrew title
write the chapter in `content/<key>.tex`, using `\linkedsection` for its sections;
copy the chapter block into `main.tex` with `chapter:<key>` as anchor and label;
foreach *section in the chapter* **do**
    add a line to the Hebrew TOC with `toc:sec:<key>` and `target:sec:<key>` (Section 4.4);
add a line for the chapter itself with `chapter:<key>`;
run `./compile_thesis.sh` and check that the merge step prints no warning;

3.3 Theorems, Figures, Tables and Definitions

Anything the Hebrew text should be able to point at gets an anchor right after its label. For a theorem, the anchor goes at the start of the body; for a figure or table, right after `\begin{figure}` or next to the `\label`.

```
\begin{restatable}{theorem}{MainTheorem}
\label{thm:main}\hypertarget{target:thm:main}{}%
Every object in the English part that carries ...
\end{restatable}

\begin{figure}[htb]
\hypertarget{target:fig:pipeline}{}%
    ...
    \label{fig:pipeline}
\end{figure}
```

The theorem above uses `restatable` from the `thmtools` package: it is stated in the introduction and restated here with `\MainTheorem*`, keeping its number, so the two statements cannot drift apart.

Theorem 1.1 *Every object in the English part that carries a named anchor can be linked to from the Hebrew pages, and the link survives any PDF viewer.*

Proof. Let x be an object with anchor `target:x`. The Hebrew source writes `\hyperlink{target:x}{...}`, which XeLaTeX turns into a link whose destination is the *name* `target:x`. Inside the Hebrew PDF alone this name is undefined. After the merge, the combined PDF contains both the link and, from the English half, the anchor of that name (Section 3.4). The merge script then replaces the name in the link by the page and position of the anchor, so the link no longer depends on the name table, which some viewers discard when they re-save a file. □

3.4 The Merge Step

`merge_thesis.py` appends the pages of `build/hebrew_pages.pdf` to `build/main.pdf`, collects the named anchors of both files, and rewrites every link that refers to a name into an explicit destination, as in the proof above. Names defined in both files are taken from the English half, except the `chapter:hebrew:*` anchors, which are taken from the Hebrew half.

The script prints how many links it resolved and, if any Hebrew link uses a name that neither document defines, a warning listing those names. A clean thesis prints no warning. The fix is to add the missing `\hypertarget` in the English source. As a last resort, the `ALIASES` table at the top of the script can map a name to one of hyperref's automatic

anchors, such as `section.2.1` or `theorem.3.1`; those names change whenever numbering changes, so prefer a real anchor.

Citations need no anchors of their own: both documents load the same `.bib` file with the same `biblatex` options, and after the merge a Hebrew citation link resolves to the entry in the English bibliography (Section 4.2). In addition, `main.tex` gives every bibliography entry an anchor named by its bib key, so `\hyperlink{Karp72}{...}` works from either side.

Chapter 4

Writing the Hebrew Pages

4.1 Right-to-Left Text and Latin Fragments

`hebrew_pages.tex` sets Hebrew as its default language with the `polyglossia` package, so ordinary paragraphs are typeset right to left and headings are aligned to the right. Inline mathematics such as `$f(k)\cdot n^{O(1)}$` works unchanged. A Latin word or phrase inside a Hebrew sentence, however, must be marked, or the bidirectional algorithm will reorder it. The template defines

```
\newcommand{\en}[1]{\beginL{\fontspec{Latin Modern Roman}#1}\endL}
```

so that `\en{NP}` or `\en{(fixed-parameter tractable)}` is typeset left to right, in a Latin font. The year on the Hebrew cover page is set the same way, with `\en{\thesisyear}`.

4.2 Citations: the `\hcite` Command

A citation inside Hebrew text is a Latin fragment too, with one more requirement: it must be a link into the English bibliography. The template's command is

```
\newcommand{\hcite}[1]{\mbox{\LR{{\fontspec{Latin Modern Roman}\cite{#1}}}}}
```

and it is used exactly like `\cite`: `\hcite{Karp72,GJ79}`. Three things make it work.

- `\LR` (from the `bidi` package) and the Latin font keep the citation label readable and left to right; `\mbox` keeps it on one line.
- `hebrew_pages.tex` loads the same `.bib` file with the same `biblatex` options as `main.tex`. The `alphabetic` style derives its labels from author names and years, so the Hebrew document prints `[Kar72]` for the same entry the English bibliography lists as `[Kar72]`, even though the Hebrew document prints no bibliography of its own.

- `biblatex` makes each citation a link to the anchor `cite.0@<key>`. In the Hebrew PDF alone that anchor is missing; after the merge it is found in the English bibliography and the link is rewired to it (Section 3.4).

One rule follows from this: an entry can only be linked to if it appears in the English bibliography, that is, if it is cited (or listed with `\nocite`) somewhere in `main.tex`. An entry cited only from Hebrew still prints its label, but the merge step reports its link as unresolved.

If you switch the English bibliography to a numeric style, switch the Hebrew document too, and keep in mind that numeric labels then depend on citation order and will differ between the two documents; the links still work, the printed numbers will not match. The alphabetic style avoids the problem.

4.3 References to English Results: the `\enref` Command

To refer to an English theorem, figure, definition or table from Hebrew text, use

```
\newcommand{\enref}[2]{\hyperlink{target:#1}{\englishTOCnumber{#2}}}
% usage, inside Hebrew text:  (<Hebrew word for Theorem> \enref{thm:main}{thm:
    main})
```

The first argument is the anchor name without its `target:` prefix, the second is the label. The command prints the object's number, read from the English build, and makes it a link to the anchor, so the Hebrew text needs no hard-coded numbers and does not break when the English part is renumbered. The Hebrew summary at the end of this document refers to Theorem 1.1, Definition 1, Figure 2.1 and Table 3.1 this way.

Behind `\enref` are the two macros that also drive the Hebrew table of contents:

```
\newcommand{\englishTOCpage}[1]{\LR{\pageref*{en-#1}}}
\newcommand{\englishTOCnumber}[1]{\LR{\ref*{en-#1}}}
```

They print a page or a number from an `en-` label, left to right, and deliberately without a link of their own (the starred forms): the enclosing `\hyperlink` is the link, and the merge script expects exactly one destination per link.

4.4 The Hebrew Table of Contents

The Hebrew TOC is written by hand, one line per English chapter or section, in `hebrew_pages.tex`. Each line combines the three ingredients above: a number, a linked Hebrew title and a page.

```
\englishTOCnumber{chapter:build}.\ \hyperlink{chapter:build}{<Hebrew chapter
    title>}
\dotfill \englishTOCpage{chapter:build}\\[0.2em]
\hspace{2em}\englishTOCnumber{toc:sec:script}\ \hyperlink{target:sec:script}{<
    Hebrew section title>}
\dotfill \englishTOCpage{toc:sec:script}\\[0.1em]
```

Unnumbered chapters (the conclusion, the bibliography) omit the number. The last two lines, for the Hebrew summary and abstract, use local labels, since those pages are in the Hebrew document itself; the English TOC reads the same labels through the `he-` prefix to list them on its side. The order of the lines, and the Hebrew titles, are yours to maintain; the numbers and pages are not.

Conclusion

This template takes care of the mechanics that make a BGU thesis awkward to typeset: an English body and a Hebrew back part in two documents, compiled by two engines, joined into a single PDF whose tables of contents, page numbers and links agree. The conventions are few. Chapters are listed in `main.tex` with an anchor and a label, sections are written with `\linkedsection`, and any theorem, figure or definition the Hebrew text refers to gets an anchor next to its label. On the Hebrew side, `\en` marks Latin fragments, `\hcite` cites, and `\enref` links to an English result and prints its number. The script compiles both parts until they agree and merges them, and its merge step tells you if a link has nowhere to go.

To turn the template into your thesis, fill in `thesis_info.tex`, replace the chapters in `content/` and the abstract and acknowledgements in `Main_pages/`, write the Hebrew summary and abstract in `hebrew_pages.tex`, and keep its table of contents in step with your chapters. Include the affidavit if your department requires it, and decide whether the thesis will be printed double-sided or read on screen, which determines whether chapters should open on odd pages. Before submission, build from a clean `build/` folder, make sure the merge step prints no warning, and open the PDF in the viewer your readers will use to click a few links in both directions.

Bibliography

[Kar72] Richard M. Karp. "Reducibility Among Combinatorial Problems". In: *Complexity of Computer Computations*. Plenum Press, 1972, pp. 85–103. DOI: 10.1007/978-1-4684-2001-2_9.

[GJ79] Michael R. Garey and David S. Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. Freeman, 1979.

[DF13] Rodney G. Downey and Michael R. Fellows. *Fundamentals of Parameterized Complexity*. Texts in Computer Science. Springer, 2013. DOI: 10.1007/978-1-4471-5559-1.

[Cyg+15] Marek Cygan, Fedor V. Fomin, Łukasz Kowalik, Daniel Lokshtanov, Dániel Marx, Marcin Pilipczuk, Michał Pilipczuk, and Saket Saurabh. *Parameterized Algorithms*. Springer, 2015. DOI: 10.1007/978-3-319-21275-3.

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תקציר

חיבור זה הוא תבנית לעבודת גמר באוניברסיטת בן-גוריון בנגב, והוא גם המדריך לשימוש בה. החלק האנגלי נכתב במסמך אחד ומקומפל עם pdfLaTeX, והעמודים העבריים, תוכן העניינים, התקציר, התמצית והכריכה העברית, נכתבים במסמך שני ומקומפלים עם XeLaTeX. הסקריפט compile_thesis.sh בונה את שניהם, חוזר על הבנייה עד ששני המסמכים קוראים זה מזה את מספרי העמודים הנכונים, ומאחד אותם לקובץ אחד (ראו איור 2.1).

ציטוטים. ציטוט בתוך טקסט עברי נכתב בפקודה \hcite בדיוק כמו \cite: לדוגמה, בעיות רבות הן NP-קשות [Kar72, GJ79], ותורת הסיבוכיות הפרמטרית [DF13, Cyg+15] מציעה דרך להתמודד איתן. התוויות בסוגריים זהות לאלו שברשימה הביבליוגרפית האנגלית, והלחיצה עליהן מובילה לערך המתאים שם.

עוגנים. כל אובייקט בחלק האנגלי שנושא עוגן בשם ...target: ניתן לקישור מכאן (הגדרה 1). הפקודה \enref מדפיסה את מספרו של האובייקט, כפי שנקרא מהבנייה האנגלית, והופכת אותו לקישור: כך, המשפט המרכזי של התבנית הוא משפט 1.1, ומוסכמות השמות מסוכמות בטבלה 3.1. המספרים בשורות אלו אינם כתובים בקובץ העברי, ולכן מספור מחדש של החלק האנגלי אינו שובר אותם.

תוכן העניינים. כל שורה בתוכן העניינים העברי בנויה משלושה רכיבים: מספר הפרק או הסעיף, הכותרת העברית כקישור, ומספר העמוד. המספר והעמוד נקראים מהבנייה האנגלית; הכותרת והסדר הם באחריות הכותב.

תמצית

מסמך זה הוא תבנית לעבודת גמר באוניברסיטת בן-גוריון בנגב, שבה חלק אנגלי וחלק עברי מופקים כקובץ אחד שהקישורים בו פועלים בשני הכיוונים. החלק האנגלי מסביר את תהליך הבנייה, את מוסכמות העוגנים והתוויות המקשרות בין שני החלקים, ואת הפקודות לכתיבת העמודים העבריים, ומדגים בדרך כל רכיב שעבודה אמיתית זקוקה לו: פרקים וסעיפים, משפט המנוסח במבוא ומוכח בהמשך, הגדרה, איור, טבלה, אלגוריתם, קטעי קוד וציטוטים. החליפו תמצית זו בתמצית של העבודה שלכם.



אוניברסיטת בן-גוריון בנגב

הפקולטה להנדסה

המחלקה להנדסת תעשייה וניהול

תבנית לעבודת גמר באוניברסיטת בן-גוריון

חיבור זה מהווה חלק מהדרישות לקבלת התואר דוקטור לפילוסופיה (PhD)

ישראל ישראלי

בהנחיית **פרופ' מקס מוסטרמן**

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